# Parametric Equations & Modeling

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**ANSWER KEY — COMPLETE WORKED SOLUTIONS**

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** **[Foundational]** • **Target: Evaluating parametric equations**

Given $x(t) = t^2 - 1$ and $y(t) = 2t + 3$, find the point on the curve at $t = 2$.

$x(2) = 4 - 1 = 3$. $y(2) = 4 + 3 = 7$. Point: $(3, 7)$.

**2** **[Foundational]** • **Target: Evaluating parametric equations**

Given $x(t) = 3\cos t$ and $y(t) = 3\sin t$, find the point at $t = \frac{\pi}{3}$, and identify the shape traced by the curve.

$x\left(\frac{\pi}{3}\right) = 3\cos\left(\frac{\pi}{3}\right) = 3\left(\frac{1}{2}\right) = \frac{3}{2}$. $y\left(\frac{\pi}{3}\right) = 3\sin\left(\frac{\pi}{3}\right) = 3\left(\frac{\sqrt{3}}{2}\right) = \frac{3\sqrt{3}}{2}$. Since $x^2 + y^2 = 9\cos^2 t + 9\sin^2 t = 9$, the curve traces a circle of radius 3 centered at the origin.

**3** **[Foundational]** • **Target: Eliminating the parameter**

Eliminate the parameter from $x(t) = t + 1$, $y(t) = t^2 - 2$ to find the rectangular equation, and identify the curve.

From $x = t + 1$: $t = x - 1$. Substitute: $y = (x - 1)^2 - 2 = x^2 - 2x + 1 - 2 = x^2 - 2x - 1$. This is a parabola.

**4** **[Foundational]** • **Target: Eliminating the parameter**

Eliminate the parameter from $x(t) = 2t - 3$, $y(t) = 4t + 1$ to find the rectangular equation.

From $x = 2t - 3$: $t = \frac{x+3}{2}$. Substitute: $y = 4\left(\frac{x+3}{2}\right) + 1 = 2(x + 3) + 1 = 2x + 7$. This is a line: $y = 2x + 7$.

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**5** [Foundational] • Target: Visualizing a parametric curve's components

The graph shows the two component functions $x(t) = t^2$ (upper curve) and $y(t) = 2t$ (lower, straight line) for the parametric curve $x(t) = t^2$, $y(t) = 2t$, plotted separately against $t$ on the horizontal axis.

**a** Eliminate the parameter $t$ to find the rectangular equation relating $x$ and $y$ for this curve. From $y = 2t$: $t = \frac{y}{2}$. Substituting into $x = t^2$: $x = \left(\frac{y}{2}\right)^2 = \frac{y^2}{4}$, so $y^2 = 4x$.

**b** Use your rectangular equation to describe the shape of the actual parametric curve (not the component graphs shown). $y^2 = 4x$ is a sideways parabola opening to the right, with vertex at the origin — very different from the two component graphs shown, which are just $x$ and $y$ plotted individually against the parameter $t$.

(a) From $y = 2t$: $t = \frac{y}{2}$. Substituting into $x = t^2$: $x = \left(\frac{y}{2}\right)^2 = \frac{y^2}{4}$, so $y^2 = 4x$.; (b) $y^2 = 4x$ is a sideways parabola opening to the right, with vertex at the origin — very different from the two component graphs shown, which are just $x$ and $y$ plotted individually against the parameter $t$.

**6** [Foundational] • Target: Parametric equations for lines and conics

Write parametric equations for the line segment from $(1, 2)$ to $(5, 10)$, using $t \in [0, 1]$.

$x(t) = 1 + 4t$, $y(t) = 2 + 8t$ for $0 \le t \le 1$ (since the change in $x$ is $5 - 1 = 4$ and the change in $y$ is $10 - 2 = 8$). Check: at $t = 0$, $(1, 2)$; at $t = 1$, $(5, 10)$ ✓.

**7** [Foundational] • Target: Parametric equations for lines and conics

Write parametric equations for an ellipse with semi-major axis 5 (horizontal) and semi-minor axis 3 (vertical), centered at the origin.

$x(t) = 5\cos t$, $y(t) = 3\sin t$, for $0 \le t < 2\pi$. Check: $\left(\frac{x}{5}\right)^2 + \left(\frac{y}{3}\right)^2 = \cos^2 t + \sin^2 t = 1$, matching the standard ellipse equation.

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Modeling projectile motion parametrically

A ball is launched with initial speed 40 m/s at an angle of $30^\circ$ above horizontal. Its position is modeled by $x(t) = 40\cos(30^\circ)\,t$ and $y(t) = 40\sin(30^\circ)\,t - 4.9t^2$. Find the ball's position at $t = 2$ seconds, to one decimal place.

$x(2) = 40\cos(30^\circ)(2) \approx 40(0.866)(2) \approx 69.3$ m. $y(2) = 40\sin(30^\circ)(2) - 4.9(4) = 40(0.5)(2) - 19.6 = 40 - 19.6 = 20.4$ m. Position: approximately $(69.3, 20.4)$.

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**9** [Intermediate] • Target: Modeling projectile motion parametrically

For the same projectile, find the time at which the ball lands (returns to $y = 0$).

Set $y(t) = 0$: $20t - 4.9t^2 = 0$, so $t(20 - 4.9t) = 0$, giving $t = 0$ or $t = \frac{20}{4.9} \approx 4.08$ seconds. The ball lands at $t \approx 4.08$ s.

**10** [Intermediate] • Target: Comparing parametric speed and direction

Two objects follow the same path $y = x^2$, but object A uses $x(t) = t$, $y(t) = t^2$ while object B uses $x(t) = 2t$, $y(t) = 4t^2$. Explain why they trace the same curve but do not move through it at the same rate.

Both satisfy $y = x^2$ (for B: $y = 4t^2 = (2t)^2 = x^2$ ✓), so they trace identical curves. But object B's $x$-coordinate changes twice as fast per unit of $t$ as object A's, meaning B moves through the same path twice as fast — the parametrization encodes speed and direction of travel, which the rectangular equation $y = x^2$ alone does not capture.

**11** [Intermediate] • Target: Restricting the parameter's domain

The parametric equations $x(t) = \cos t$, $y(t) = \sin t$ trace the full unit circle for $t \in [0, 2\pi)$. Describe the portion of the circle traced when the domain is restricted to $t \in [0, \frac{\pi}{2}]$.

Restricting to $t \in [0, \frac{\pi}{2}]$ traces only the arc from $(1, 0)$ (at $t = 0$) to $(0, 1)$ (at $t = \frac{\pi}{2}$) — the quarter of the circle lying in Quadrant I.

**12** [Intermediate] • Target: Restricting the parameter's domain

Explain why restricting $t$ in $x(t) = t^2$, $y(t) = t$ to $t \geq 0$ changes the rectangular relation from $x = y^2$ (a full sideways parabola) to only its upper half.

Eliminating the parameter gives $x = y^2$ for all $t$, but since $y = t$ directly, restricting $t \geq 0$ forces $y \geq 0$ as well — so only the upper half of the sideways parabola (where $y \geq 0$) is actually traced, even though the unrestricted equation $x = y^2$ would include the lower half too.

**13** [Intermediate] • Target: Parametric equations for circular motion with a phase shift

A point moves along a circle of radius 4 according to $x(t) = 4\cos\left(t + \frac{\pi}{4}\right)$, $y(t) = 4\sin\left(t + \frac{\pi}{4}\right)$. Find its position at $t = 0$, and explain the effect of the $\frac{\pi}{4}$ phase shift compared to $x(t) = 4\cos t$, $y(t) = 4\sin t$.

At $t = 0$: $x(0) = 4\cos\left(\frac{\pi}{4}\right) = 4\left(\frac{\sqrt{2}}{2}\right) = 2\sqrt{2}$, $y(0) = 4\sin\left(\frac{\pi}{4}\right) = 2\sqrt{2}$. Without the phase shift, the point would start at $(4, 0)$ when $t = 0$; the added $\frac{\pi}{4}$ advances the starting position by an eighth of a full revolution counterclockwise, to $(2\sqrt{2}, 2\sqrt{2})$, while the circle traced and the speed of travel remain unchanged.

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**14** [Intermediate] • Target: Comparing parametric speed and direction

Eliminate the parameter to determine the Cartesian equation of the curve $x = 1 + 2\sec t$, $y = -2 + 3\tan t$. State the shape, center, and vertices.

Isolate the trigonometric terms: $\sec t = \frac{x-1}{2}, \quad \tan t = \frac{y+2}{3}$. Apply the fundamental Pythagorean identity $\sec^2 t - \tan^2 t = 1$: $\left(\frac{x-1}{2}\right)^2 - \left(\frac{y+2}{3}\right)^2 = 1 \Rightarrow \frac{(x-1)^2}{4} - \frac{(y+2)^2}{9} = 1$. This is a horizontal hyperbola with:Center: $(h, k) = (1, -2)$Transverse semi-axis: $a = 2$Vertices: $(h \pm a, k) = (1 \pm 2, -2) = (-1, -2)$ and $(3, -2)$

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** [Challenging] • Target: Restricting the parameter's domain • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

Given vectors $\mathbf{a} = 2\mathbf{i} - 5\mathbf{j}$ and $\mathbf{b} = -3\mathbf{i} + 4\mathbf{j}$, find the scalar projection of $\mathbf{a}$ onto $\mathbf{b}$ and the angle $\theta$ between them to the nearest tenth of a degree.

Calculate the dot product: $\mathbf{a} \cdot \mathbf{b} = (2)(-3) + (-5)(4) = -6 - 20 = -26$. Calculate the magnitudes: $|\mathbf{a}| = \sqrt{2^2 + (-5)^2} = \sqrt{4 + 25} = \sqrt{29} \approx 5.385$, $|\mathbf{b}| = \sqrt{(-3)^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$. Scalar projection of $\mathbf{a}$ onto $\mathbf{b}$: $\text{comp}_{\mathbf{b}}\mathbf{a} = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{b}|} = \frac{-26}{5} = -5.2$. Angle $\theta$ between vectors.

**16** [Challenging] • Target: Parametric equations for circular motion with a phase shift • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

[Vectors, Lines & Parametric Kinematics · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input $t$ when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

**17** [Challenging] • Target: Evaluating parametric equations • Cross-Lesson: 4.01 Inverse & Composite Functions

Decompose the vector $\mathbf{u} = \langle 6, 2 \rangle$ into components parallel and orthogonal to $\mathbf{v} = \langle 3, 4 \rangle$.

Step 1: Compute dot product and squared magnitude: $\mathbf{u} \cdot \mathbf{v} = (6)(3) + (2)(4) = 18 + 8 = 26$. $||\mathbf{v}||^2 = 3^2 + 4^2 = 9 + 16 = 25$. Step 2: Vector projection (parallel component): $\mathbf{u}_{\parallel} = \text{proj}_{\mathbf{v}}\mathbf{u} = \left(\frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{v}||^2}\right)\mathbf{v} = \frac{26}{25}\langle 3, 4 \rangle = \left\langle \frac{78}{25}, \frac{104}{25} \right\rangle = \langle 3.12, 4.16 \rangle$. Step 3: Orthogonal component.

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**18** [Challenging] • Target: Eliminating the parameter • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

A right circular cone has a base radius of 6 cm and a slant height of 10 cm. Find (a) its perpendicular vertical height $h$, (b) its volume $V$ in terms of $\pi$, and (c) its total surface area in terms of $\pi$.

(a) Vertical height: Using the Pythagorean relation $l^2 = r^2 + h^2$:
$h = \sqrt{10^2 - 6^2} = \sqrt{100 - 36} = \sqrt{64} = \mathbf{8}$ cm (b) Volume:

$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(6^2)(8) = \frac{1}{3}\pi(36)(8) = 12 \times 8\pi = \mathbf{96\pi}$ cm³ $\approx 301.59$ cm³ (c) Total surface area:

$\text{TSA} = \pi r^2 + \pi r l = \pi(6^2) + \pi(6)(10) = 36\pi + 60\pi = \mathbf{96\pi}$ cm² $\approx 301.59$ cm²

**19** [Challenging] • Target: Visualizing a parametric curve's components • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

Find all solutions to the trigonometric equation $2\sin^2 x - 3\sin x + 1 = 0$ on the domain $x \in [0, 2\pi)$.

Step 1: Factor the quadratic in $\sin x$: $(2\sin x - 1)(\sin x - 1) = 0$. Step 2: Solve each factor branch:Branch 1: $2\sin x - 1 = 0 \Rightarrow \sin x = \frac{1}{2}$. On $[0, 2\pi)$, $x = \frac{\pi}{6}$ (Quadrant I) or $x = \frac{5\pi}{6}$ (Quadrant II).Branch 2: $\sin x - 1 = 0 \Rightarrow \sin x = 1$. On $[0, 2\pi)$, $x = \frac{\pi}{2}$.The complete solution set is $x \in \left\{\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}\right\}$.

**20** [Challenging] • Target: Parametric equations for lines and conics • Cross-Lesson: 1.02 Rational Functions & Asymptotes

Find the volume of a right circular cone with base radius 6 and height 8.

Apply the cone volume formula $V = \frac{1}{3}\pi r^2 h$: $V = \frac{1}{3}\pi(6^2)(8) = \frac{1}{3}\pi(36)(8) = 12 \times 8\pi = \mathbf{96\pi} \approx 301.59$ Key Mathematical Principle & Verification: In exam revision, always verify that the final value satisfies the domain constraints, units, and original governing equation by direct back-substitution into the prompt.

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## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Evaluating parametric equations

Eliminate the parameter from $x(t) = t - 2$ and $y(t) = -3t + 4$ to find the rectangular equation for $y$ in terms of $x$.

**a.** $y = -3x - 1$

**b.** $y = -3x + 4$

**c.** $y = 3x - 2$

**d.** $y = -3x - 2$ ✓

From $x = t - 2$: $t = x + 2$. Substitute: $y = -3(x + 2) + 4 = -3x - 2$.

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**22** [Foundational] • Target: Eliminating the parameter

A curve is given by $x(t) = 4\cos t$, $y(t) = 4\sin t$. Find the exact point on the curve at $t = \frac{11\pi}{6}$ $(330^\circ)$.

**a.** $(2\sqrt{3}, 2)$

**b.** $(2\sqrt{3}, -2)$ ✓

**c.** $(-2, 2\sqrt{3})$

**d.** $(-2\sqrt{3}, -2)$

$x = 4\cos\left(\frac{11\pi}{6}\right) = 2\sqrt{3}$. $y = 4\sin\left(\frac{11\pi}{6}\right) = -2$. Point: $(2\sqrt{3}, -2)$.

**23** [Foundational] • Target: Visualizing a parametric curve's components

A projectile is launched with initial speed 34 m/s at $30^\circ$ above horizontal, so $y(t) = 34\sin(30^\circ)\,t - 4.9t^2$. Find $y(2)$, to one decimal place.

**a.** 12.4

**b.** 19.4

**c.** 16.4

**d.** 14.4 ✓

$y(2) = 34\sin(30^\circ)(2) - 4.9(2)^2 \approx 34(0.5)(2) - 4.9(4) \approx 14.4$ m.

**24** [Foundational] • Target: Parametric equations for lines and conics

Two particles, A and B, move in the xy-plane. Their position vectors at time $t$ seconds, for $t \geq 0$, are given by $\mathbf{r}_A(t) = (2t;\, 4t - 1)$ and $\mathbf{r}_B(t) = (3 + 5t;\, 1 - 2t)$ respectively. The particles start their motion at $t = 0$. Find the minimum distance between particle A and particle B.

**a.** $\sqrt{26}$ ✓

**b.** $2\sqrt{13}$

**c.** $3\sqrt{10}$

**d.** $4\sqrt{5}$

The position vectors of particle A and particle B at time $t$ are given by $\mathbf{r}_A(t) = (2t;\, 4t - 1)$ and $\mathbf{r}_B(t) = (3 + 5t;\, 1 - 2t)$. The vector representing the displacement from particle B to particle A at time $t$ is $\mathbf{d}(t) = \mathbf{r}_A(t) - \mathbf{r}_B(t) = (2t - (3 + 5t);\, (4t - 1) - (1 - 2t)) = (-3t - 3;\, 6t - 2)$. The distance between the particles at time $t$ is the magnitude of this displacement vector.

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**25** [Foundational] • Target: Modeling projectile motion parametrically

Consider two vectors, $\mathbf{u} = (3; k)$ and $\mathbf{v} = (-2; 4)$. If $\mathbf{u}$ is parallel to $\mathbf{v}$, find the value of the scalar parameter $k$.

**a.** **-6 ✓**

**b.** 6

**c.** 3

**d.** -3

Two vectors are parallel if one is a scalar multiple of the other. Therefore, there must exist a scalar $c$ such that $\mathbf{u} = c\mathbf{v}$. This gives us the equations: $3 = c(-2)$ $k = c(4)$ From the first equation, we can solve for $c$: $c = \frac{3}{-2} = -\frac{3}{2}$ Now substitute this value of $c$ into the second equation to find $k$: $k = \left(-\frac{3}{2}\right)(4)$ $k = -\frac{12}{2}$ $k = -6$ Therefore, the value of $k$ is -6. A common algebraic trap is to mistakenly set the components equal with.

**26** [Foundational] • Target: Comparing parametric speed and direction

Two vectors are given by $\vec{u} = 3\hat{i} + k\hat{j}$ and $\vec{v} = -\hat{i} + 4\hat{j}$. If $\vec{u}$ and $\vec{v}$ are orthogonal, find the value of the scalar parameter $k$.

**a.** 3/4

**b.** **4/3 ✓**

**c.** -3/4

**d.** -4/3

Two vectors $\vec{u}$ and $\vec{v}$ are orthogonal if and only if their dot product is zero. The dot product of $\vec{u} = u_1\hat{i} + u_2\hat{j}$ and $\vec{v} = v_1\hat{i} + v_2\hat{j}$ is given by $\vec{u} \cdot \vec{v} = u_1v_1 + u_2v_2$. Given $\vec{u} = 3\hat{i} + k\hat{j}$ and $\vec{v} = -\hat{i} + 4\hat{j}$, their dot product is: $\vec{u} \cdot \vec{v} = (3)(-1) + (k)(4)$ $\vec{u} \cdot \vec{v} = -3 + 4k$ For the vectors to be orthogonal, their dot product must be zero: $-3 + 4k = 0$ $4k = 3$.

**27** [Foundational] • Target: Restricting the parameter's domain

Consider the vector $\mathbf{v}$ with initial point $P = (2, 3)$ and terminal point $Q = (5, 7)$. What are the components of the vector $\mathbf{v}$?

**a.** **■3, 4■ ✓**

**b.** ■7, 10■

**c.** ■-3, -4■

**d.** ■4, 3■

The components of a vector from point $P = (x_1, y_1)$ to point $Q = (x_2, y_2)$ are given by $\langle x_2 - x_1, y_2 - y_1 \rangle$. In this case, $P = (2, 3)$ and $Q = (5, 7)$. Therefore, the components of vector $\mathbf{v}$ are $\langle 5 - 2, 7 - 3 \rangle = \langle 3, 4 \rangle$.

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**28** [Foundational] • Target: Parametric equations for circular motion with a phase shift

Find the unit vector in the direction of $\mathbf{v} = \langle -8, 15 \rangle$.

**a.** $\left\langle \frac{9}{17}, \frac{15}{17} \right\rangle$

**b.** $\left\langle \frac{15}{17}, -\frac{8}{17} \right\rangle$

**c.** $\left\langle -\frac{8}{17}, \frac{15}{17} \right\rangle$ ✓

**d.** $\langle -8, 15 \rangle$

$|\mathbf{v}| = 17$. Unit vector $= \frac{\mathbf{v}}{|\mathbf{v}|} = \left\langle \frac{-8}{17}, \frac{15}{17} \right\rangle$.

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Evaluating parametric equations

In the standard $(x, y)$ coordinate plane, the lines with equations $2x - y = 7$ and $x + y = 5$ intersect at the point $(a, b)$. What is the value of $a$?

**a.** 1

**b.** 3

**c.** **4** ✓

**d.** 5

The point of intersection $(a, b)$ satisfies both linear equations simultaneously: 1) $2a - b = 7$ 2) $a + b = 5$ To solve for $a$, add the two equations together to eliminate $b$: $(2a - b) + (a + b) = 7 + 5$ $3a = 12$ $a = 4$ Substituting $a = 4$ into the second equation gives $4 + b = 5$, which yields $b = 1$. Thus, the lines intersect at $(4, 1)$, and the value of $a$ is $4$. - A common misconception is solving for $b = 1$ and selecting 1 (Choice A). - Another error is an arithmetic mistake in dividing 12 by 3.

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**30** [Intermediate] • Target: Eliminating the parameter

Consider two points in the Cartesian plane, $P = (2, 3)$ and $Q = (5, 7)$. What is the component form of the vector $\vec{PQ}$?

**a.** $\langle 3, 4 \rangle$ ✓

**b.** $\langle -3, -4 \rangle$

**c.** $\langle 7, 10 \rangle$

**d.** $\langle 1, 4 \rangle$

To find the component form of a vector $\vec{PQ}$ from point $P(x_1, y_1)$ to point $Q(x_2, y_2)$, we subtract the coordinates of the initial point $P$ from the coordinates of the terminal point $Q$. The formula is $\vec{PQ} = \langle x_2 - x_1, y_2 - y_1 \rangle$. Given $P = (2, 3)$ and $Q = (5, 7)$, we have $x_1 = 2$, $y_1 = 3$, $x_2 = 5$, and $y_2 = 7$. Applying the formula: $\vec{PQ} = \langle 5 - 2, 7 - 3 \rangle$ $\vec{PQ} = \langle 3, 4 \rangle$ Therefore, the component form of the vector $\vec{PQ}$ is.

**31** [Intermediate] • Target: Visualizing a parametric curve's components

Consider the vector $\mathbf{v} = \langle 3, 4 \rangle$. What is the magnitude of this vector?

**a.** **5** ✓

**b.** 7

**c.** 12

**d.** 25

The magnitude of a vector $\mathbf{v} = \langle x, y \rangle$ is given by the formula $|\mathbf{v}| = \sqrt{x^2 + y^2}$. For the given vector $\mathbf{v} = \langle 3, 4 \rangle$, we have x = 3 and y = 4. Substituting these values into the formula: $|\mathbf{v}| = \sqrt{3^2 + 4^2}$ $|\mathbf{v}| = \sqrt{9 + 16}$ $|\mathbf{v}| = \sqrt{25}$ $|\mathbf{v}| = 5$ Therefore, the magnitude of the vector $\mathbf{v}$ is 5.

**32** [Intermediate] • Target: Parametric equations for lines and conics

Two lines have slopes $m_1 = -1$, $m_2 = -1$ and y-intercepts $b_1 = -2$, $b_2 = -2$. Classify the system.

**a.** Consistent and independent (one solution)

**b.** **Consistent and dependent (infinitely many solutions)** ✓

**c.** Inconsistent (no solution)

**d.** Cannot be determined

Different slopes give exactly one intersection (independent). Equal slopes with equal intercepts give the same line (dependent, infinite solutions). Equal slopes with different intercepts give parallel lines (inconsistent, no solution).

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**33** [Intermediate] • Target: Modeling projectile motion parametrically

Find the projection of $\mathbf{u} = \langle 3, 2 \rangle$ onto $\mathbf{v} = \langle 2, 2 \rangle$.

**a.** $\left\langle \frac{3}{2}, \frac{5}{2} \right\rangle$

**b.** $\left\langle \frac{5}{2}, \frac{5}{2} \right\rangle$ ✓

**c.** $\left\langle \frac{7}{2}, \frac{5}{2} \right\rangle$

**d.** $\frac{5}{4}$

$\text{proj}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|^2}\mathbf{v} = \frac{10}{8}\langle 2, 2 \rangle = \left\langle \frac{5}{2}, \frac{5}{2} \right\rangle$.

**34** [Intermediate] • Target: Comparing parametric speed and direction

Find the angle between $\mathbf{u} = \langle 2, 1 \rangle$ and $\mathbf{v} = \langle 4, 2 \rangle$ (to one decimal place).

**a.** **0.0°** ✓

**b.** 20.0°

**c.** 180.0°

**d.** 10.0°

$\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|} = \frac{10}{2.24 \times 4.47} \approx 1.0$, so $\theta \approx 0.0°$.

**35** [Intermediate] • Target: Restricting the parameter's domain

Find the parametric equations for the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$.

**a.** $x = 4\sin t, y = 3\cos t$

**b.** $x = 4t, y = 3t$

**c.** $x = 4\cos t, y = 3\sin t$ ✓

**d.** $x = \cos t, y = \sin t$

Applying Conic Sections principles, Standard parametrization. Therefore, the verified correct option is $x = 4\cos t, y = 3\sin t$.

**36** [Intermediate] • Target: Parametric equations for circular motion with a phase shift

For the hyperbola $\frac{x^2}{4} - \frac{y^2}{9} = 1$, what are the equations of the asymptotes?

**a.** $y = \pm\frac{3}{2}x$ ✓

**b.** $y = \pm\frac{2}{3}x$

**c.** $y = \pm 6x$

**d.** $y = \pm\frac{5}{2}x$

Asymptotes of $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ are $y = \pm\frac{b}{a}x = \pm\frac{3}{2}x$.

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**37** [Intermediate] • Target: Evaluating parametric equations

Find the period and phase shift of $y = \tan(1/2x - 1)$.

**a.** Period $2\pi$, phase shift 0

**b.** Period $2\pi$, phase shift 2 units left

**c.** Period $\pi$, phase shift 2 units right

**d.** **Period $2\pi$, phase shift 2 units right ✓**

For $\tan(Bx - C)$: period $= \frac{\pi}{B} = 2\pi$, phase shift $= \frac{C}{B} = 2$ units right.

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## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Eliminating the parameter • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

Graph $y = 2\sin(3x - \pi)$: state amplitude, period, phase shift.

**a.** Amp 2, period $2\pi$, phase $\pi$

**b.** Amp 2, period $\pi$

**c.** Amp 3, period $2\pi/3$

**d.** **Amplitude 2, period $2\pi/3$, phase shift $\pi/3$ right ✓**

$A = 2, B = 3$: period $= 2\pi/3$. Phase: $3x - \pi = 0 \Rightarrow x = \pi/3$.

**39** [Challenging] • Target: Visualizing a parametric curve's components • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

What is the center of the circle $(x - 3)^2 + (y + 2)^2 = 16$?

**a.** (-3, 2)

**b.** **(3, -2) ✓**

**c.** (3, 2)

**d.** (-2, 3)

Comparing to $(x - h)^2 + (y - k)^2 = r^2$: center $(h, k) = (3, -2)$.

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**40** [Challenging] • Target: Parametric equations for lines and conics • Cross-Lesson: 1.02 Rational Functions & Asymptotes

Let $m$ be a real parameter. Consider the quadratic equation $z^2 - 2mz + (m^2 + 3m - 6) = 0$ with complex roots $z_1$ and $z_2$. In the complex plane, the origin $O(0)$ and the points representing $z_1$ and $z_2$ form the vertices of a non-degenerate equilateral triangle. What is the sum of the areas of all possible equilateral triangles that can be formed in this way?

**a.** $15\sqrt{3}$ ✓

**b.** $12\sqrt{3}$

**c.** $\frac{15\sqrt{3}}{2}$

**d.** $9\sqrt{3}$

We solve the problem by synthesizing Vieta's formulas with the geometric properties of complex numbers. **Step 1: Condition for an Equilateral Triangle with the Origin** In the complex plane, three points $0$, $z_1$, and $z_2$ form the vertices of an equilateral triangle if and only if one non-zero vertex is obtained by rotating the other about the origin by $\pm\frac{\pi}{3}$: $\frac{z_2}{z_1} = e^{\pm i\pi/3} = \frac{1 \pm i\sqrt{3}}{2}$ This is equivalent to the quadratic in $\frac{z_2}{z_1}$.

**41** [Challenging] • Target: Modeling projectile motion parametrically • Cross-Lesson: 1.03 Rates of Change & Concavity

Consider the system of linear equations represented by the matrix equation $(1, 1, 1; 1, 2, 3; 3, 5, k)(x; y; z) = (6; 14; m)$ where $k$ and $m$ are real constants. Suppose the system is consistent and has infinitely many solutions. Let $P(x, y, z)$ be the unique solution that minimizes the squared Euclidean distance from the origin, $d^2 = x^2 + y^2 + z^2$. What is the value of $k + m + d^2$?

**a.** **55** ✓

**b.** 47

**c.** 38

**d.** 109

We solve the problem in three main stages: 1. **Determine the parameter $k$:** For the system to have infinitely many solutions, the coefficient matrix must be singular (its determinant must be zero): $\det(1, 1, 1; 1, 2, 3; 3, 5, k) = 1(2k - 15) - 1(k - 9) + 1(5 - 6) = k - 7$ Setting $k - 7 = 0$ yields $k = 7$. 2. **Determine the parameter $m$:** Set up the augmented matrix with $k = 7$ and perform Gaussian elimination.

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**42** [Challenging] • Target: Comparing parametric speed and direction • Cross-Lesson: 2.01 Exponential Functions & Growth Models

Consider the system of linear equations below: $\{3x - 2y = 8; kx + 6y = 5$ For which of the following values of the constant $k$ does the system have no solution $(x, y)$?

**a.** **-9 ✓**

**b.** -4

**c.** 4

**d.** 9

A system of two linear equations in two variables has no solution if and only if the lines represented by the equations are parallel and distinct (they have the same slope but different $y$-intercepts). First, rewrite each equation in slope-intercept form $y = mx + b$: 1. For the first equation:
$3x - 2y = 8 \Rightarrow -2y = -3x + 8 \Rightarrow y = \frac{3}{2}x - 4$ The slope of the first line is $m_1 = \frac{3}{2}$ and the $y$-intercept is $-4$. 2. For the second equation.

**43** [Challenging] • Target: Restricting the parameter's domain • Cross-Lesson: 2.02 Logarithmic Functions & Properties

If the real numbers $x$ and $y$ satisfy the system of equations $x^3 - 3xy^2, = 10; 3x^2y - y^3, = 5$ what is the value of $x^2 + y^2$?

**a.** $\sqrt{5}$

**b.** 5 ✓

**c.** $5\sqrt{5}$

**d.** 25

We can solve this system using algebraic expansion or complex numbers. **Method 1: Squaring and Adding** Square both equations and add them together: $(x^3 - 3xy^2)^2 + (3x^2y - y^3)^2 = 10^2 + 5^2 = 100 + 25 = 125$ Expanding the left-hand side: $(x^6 - 6x^4y^2 + 9x^2y^4) + (9x^4y^2 - 6x^2y^4 + y^6) = x^6 + 3x^4y^2 + 3x^2y^4 + y^6$ Notice that this expression is the binomial expansion of $(x^2 + y^2)^3$: $(x^2 + y^2)^3 = 125$ Taking the real cube root of both sides gives: $x^2 + y^2 = \sqrt[3]{125} = 5$ **Method 2.

**44** [Challenging] • Target: Parametric equations for circular motion with a phase shift • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

In the system of linear equations below, $k$ is a real constant. $(k - 1)x + 6y, = 18; 2x + ky, = 12$ For which value of $k$ does the system have infinitely many solutions $(x, y)$?

**a.** $-4$

**b.** $-3$

**c.** 3

**d.** 4 ✓

For a system of two linear equations in two variables to have infinitely many solutions, the two equations must represent the exact same line. This requires the ratios of the corresponding coefficients and the constant terms to all be equal: $\frac{k-1}{2} = \frac{6}{k} = \frac{18}{12}$ First, simplify the ratio of the constant terms: $\frac{18}{12} = \frac{3}{2}$ Next, set the ratio of the $y$-coefficients equal to $\frac{3}{2}$: $\frac{6}{k} = \frac{3}{2} \Rightarrow 3k = 12 \Rightarrow k = 4$ Verify this value.

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**45** [Challenging] • Target: Evaluating parametric equations • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

In the standard $(x, y)$ coordinate plane, the system of equations $\{y = |x^2 - 4|; y = 2x + k$ has exactly 3 distinct real solutions $(x, y)$. What is the sum of all possible values of the real constant $k$?

**a.** 4

**b.** 5

**c.** **9** ✓

**d.** 11

The system represents the intersections of the absolute-value quadratic graph $y = |x^2 - 4|$ and the family of parallel lines $y = 2x + k$ of slope 2. The piecewise definition of $y = |x^2 - 4|$ is: \[ y = \begin{cases} x^2 - 4, & \text{if } |x| \ge 2 \\ 4 - x^2, & \text{if } |x| 2 \) and $1 - \sqrt{10}5$: 2 solutions (outer branches only). - For $4 < k < 5$: 4 solutions (2 on the inverted arch, 2 on the outer branches). - For $-4 < k < 4$: 2 solutions. - For $k = -4$: 1 solution (passing through $(2, 0)$). - For.

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